

SPARSH PATEL

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING

Master of Science in Financial Engineering

GPA: 3.8/4.0

SRM IST, DEPARTMENT OF COMPUTING TECHNOLOGIES

Bachelor of Technology in Computer Science Engineering

GPA: 9.2/10.0

Brooklyn, NY

Expected 05/27

Chennai, IN

10/24

TECHNICAL SKILLS / CERTIFICATIONS

- Skills: Python, SQL, Excel, PowerBI, R, C/C++
- Certifications: Google Data Analytics Professional Certificate, Quant Finance Institute Certified, Bloomberg Market Concepts

COURSEWORK HIGHLIGHTS

- Mathematics & Statistics: Linear Algebra, Advanced Calculus and Complex Analysis, Probability and Queueing Theory, Discrete Mathematics
- Programming: Design of Algorithms, Data Mining and Analytics, Artificial Intelligence, ML in Financial Engineering
- Finance and Economics: Quantitative Methods in Finance, Introduction to Derivative Securities

EXPERIENCE

VETA INVESTMENT PARTNERS, Princeton, NJ

05/26-08/26

Quant Research Intern

- Built a ~20-module Python options analytics pipeline integrating reconciled custodian data in PostgreSQL, portfolio risk analytics, hedge recommendations and strategy pricing with automated nightly Greeks, returns and advisor reporting
- Developed a dividend-adjusted Black-Scholes-Merton pricing and Greeks engine with implied-volatility solving; validated Greeks to ~1e-7 against an independent library and reconciled strategy prices to the cent against the firm workbook
- Engineered pricing and recommendation models across 5 option-strategy families, using option-chain midpoints and numerical zero-cost collar solving to price delta-sized hedges for concentrated portfolios, net of existing protection

SHAH INVESTORS HOME LTD, Ahmedabad, GJ

01/25 - 05/25

Quant Trading Analyst Intern

- Formulated data-driven option selling strategies using a hybrid of Black-Scholes valuation, Monte Carlo simulation and Greeks, optimizing premium selection and avoiding two instances of major losses
- Boosted revenue by 4% over a three-month period by analysing brokerage trends and customer segments via Power BI; cut messaging costs by 40% by evaluating platform performance and streamlining communication workflows
- Engineered a bull-bear trading strategy for algorithmic trading through Python and AceEquity data; conducted historical back testing to improve signal accuracy and improve strategy robustness and execution time by 11%

CRM MESSAGING, Delaware (Remote)

07/24 - 08/24

Quant Analyst Intern

- Applied NLP-based feature extraction on historical customer chats combined with quantitative engagement metrics to improve message targeting reach through feature-based targeting. Improving reach by 9%
- Optimized system performance metrics through statistical analysis of bottlenecks, boosting organic traffic by 6%

RESEARCH / FINANCIAL MODELLING

INTERNATIONAL ASSOCIATION OF QUANTITATIVE FINANCE, New York, US

01/26 - 02/26

2026 IAQF Student Competition — Winning Paper (Team Sharpe Minds)

- “Pegged... Until It’s Not: Stablecoin Risk and Market Dislocation.”
- Studied BTC/USD vs. BTC/stablecoin pricing across Coinbase and Kraken using minute-level data around the March 2023 SVB crisis; decomposed cross-currency basis into peg deviation and exchange-level friction and quantified a persistent post-crisis dislocation and flight-to-quality from USDC to USDT

COMPOUND OPTION PRICING & SENSITIVITY ANALYSIS, New York, US

11/25 - 12/25

NYU Tandon, Department of Finance & Risk Engineering

- Developed a compound option (option-on-option) pricing framework under Geometric Brownian Motion (GBM) using Monte Carlo simulation in Python, implementing risk-neutral valuation and analyzing simulated payoff distributions
- Conducted sensitivity, Greek, and scenario analysis by varying volatility, moneyness, and inner/outer maturities; computed Greeks via finite differences and visualized non-linear price, convexity, and hedging risk behaviour

STOCHASTIC INTEREST RATE MODELLING, Ahmedabad, India

02/25 - 03/25

- Developed and implemented the Vasicek and Cox-Ingersoll-Ross (CIR) stochastic interest rate models, capturing mean-reverting behaviour in interest rate dynamics to reflect realistic economic conditions
- Performed market calibration using Maximum Likelihood Estimation (MLE) to estimate key parameters such as speed of mean reversion, long-term mean, and volatility, ensuring model alignment with observed interest rate movements

ENHANCING MEDICAL DECISION SUPPORT USING GPT, Chennai, India

07/23 - 12/23

SRM IST, Department of Computing Technologies

- Received best paper award in the Futuristic Computing Trends in Financial Technologies (FCOM-Fintech’24) conference
- Utilized LoRA (Low Rank Adaptation) to fine-tune a Hugging Face Transformer on a doctor-patient conversation dataset, greatly reducing training time and GPU memory usage while maintaining accurate task-specific performance